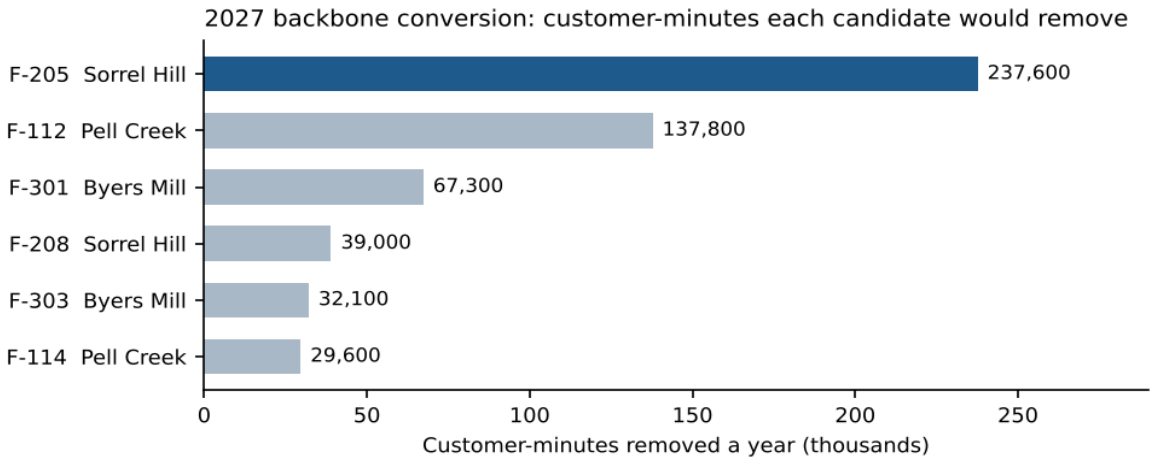
# 2027 backbone conversion

Memo for the 2027 Storm Hardening Plan filing, due to the commission 1 December 2026. October 2026.

## Recommendation

Convert the backbone of **F-205 (Sorrel Hill)** in 2027. Valued the way the plan values every project, the conversion removes **237,600 customer-minutes** of sustained interruption a year, worth **USD 499,000 a year** at the commission's incentive of USD 2.10 a customer-minute. The next candidate, F-112 (Pell Creek), removes 137,800.

| Feeder | Substation | Backbone miles | Estimated cost (USD million) | Customer-minutes removed a year |
|---|---|---|---|---|
| F-205 | Sorrel Hill | 14.4 | 6.0 | 237,600 |
| F-112 | Pell Creek | 12.6 | 5.3 | 137,800 |
| F-301 | Byers Mill | 14.2 | 6.0 | 67,300 |
| F-208 | Sorrel Hill | 12.4 | 5.2 | 39,000 |
| F-303 | Byers Mill | 10.5 | 4.4 | 32,100 |
| F-114 | Pell Creek | 11.2 | 4.7 | 29,600 |




## Why F-112 is not first

Pell Creek's three feeders (F-112, F-114 and F-117) opened their breakers in the same minute 12 times outside the major event days from 2019 to 2025. Each time the outage system logged the interruption on each feeder's own breaker, with the cause the crew found on the line that supplies the substation. A fault on one feeder's backbone does not take its sibling feeders' customers off, so these are losses of the substation's supply, and converting a feeder's backbone does not prevent them. Credited as faults on the backbone they would put F-112 at 455,400 customer-minutes a year and F-114 at 285,400; without them F-112 removes 137,800 and F-114 29,600. Byers Mill's feeders share 3 such minutes; Sorrel Hill's share none.

## How the figures were built

Both outage exports for 2019 to 2025: the legacy feeder numbers and cause codes mapped to the current system's, its UTC stamps put on Eastern time. Tickets that began on a major event day the commission accepted are left out. Of the rest, the feeder breaker and line recloser tickets caused by vegetation, wind or wildlife, less the tickets every feeder at the substation opened in the same minute; customer-minutes are customers interrupted times minutes to restoration; nine in ten of them, divided by seven. Each candidate's storm record (the accepted major event days of 2023 to 2025) and its taps' record are in hardening_tables_2027.xlsx.